

The Schur error of the small-volume reduction is an ultraviolet problem: a fixed-lattice theorem is available, the renormalized regime is not

Inequality (H2) of the Feshbach reduction is estimated here at second order with the fibered vacuum, on the abelian valley and with the cutoff Λ kept explicit. Two terms have different characters. The non-adiabatic (Berry) coupling, produced by the constant-mode momentum acting on the a -dependent Gaussian vacuum, has Schur error

$$\eta_{\text{Berry}} \simeq \frac{g^{8/3}}{96\pi^2} \log(\Lambda L) \langle |\partial_\xi \psi|^2 \rangle \frac{\hbar c}{L},$$

of relative order $g^2 \log(\Lambda L)$ to the zero-mode gap $g^{2/3} \hbar c/L$: logarithmic, and of the size and form of the one-loop renormalization of the coupling, so it is absorbed by writing the gap in terms of $g(L)$. The cubic coupling W_3 produces at second order a constant vacuum shift $\eta_0 \sim g^2 (\hbar c/L) (\Lambda L)^4$ and an energy derivative $d\eta_0/dE \sim g^2 (\Lambda L)^2$; the shift cancels from the gap, but the derivative rescales it, and the transfer lemma holds only when $g^2 (\Lambda L)^2 \ll 1$, that is $g \ll 1/N_s$ on a lattice of N_s sites per side. Consequently the reduction proves the small-volume gap $\Delta \geq \delta_1^{(3)} g^{2/3} (1 - o(1)) \hbar c/L$ as a fixed-lattice theorem, in the regime $g \rightarrow 0$ at fixed N_s , and does not reach the regime $g^2 \log N_s \ll 1$ of the renormalized expansion, where the bare Gaussian fiber is far from the true nonzero-mode vacuum. The small-volume corner is therefore an ultraviolet problem at fixed L : its control uniformly in the cutoff is a renormalization theorem, the same obligation as T4 of the obligations map. The route's step 1 splits accordingly into a provable fixed-lattice statement and a renormalized statement that needs vacuum dressing. Constants explicit throughout; nothing here is promoted.

1. The refined transfer lemma with a constant shift

Lemma 1 of the Feshbach note bounds the Schur term $S(E) = B(D - E)^{-1} B^*$ by a constant. When $S(E)$ contains a large part proportional to the identity, that part shifts all eigenvalues of the compression equally and must not be charged to the gap.

Lemma 1'. In the setting of Lemma 1 suppose that for $E \in I = [a_0 - \eta_1 - \bar{\eta}_0, a_1]$,

$$S(E) = \eta_0(E) 1 + R(E), \quad 0 \leq R(E) \leq \epsilon(A - a_0) + \eta_1,$$

with η_0 a nonnegative nondecreasing function, $\bar{\eta}_0 = \sup_I \eta_0$, $0 \leq \epsilon < 1$, and $\eta_0(E)$ differentiable with $0 \leq \eta'_0(E) \leq \theta < 1$ on I . Then

$$E_1 - E_0 \geq \frac{(1 - \epsilon)(a_1 - a_0) - 2\eta_1}{1 + \theta}.$$

Proof. $F(E) = A - E - \eta_0(E) - R(E)$. The count of negative eigenvalues of $F(E)$ lies between the counts of $A - E - \eta_0(E) - \epsilon(A - a_0) - \eta_1$ and $A - E - \eta_0(E)$. Write $\tilde{E} = E + \eta_0(E)$, which is increasing in E with $d\tilde{E}/dE \leq 1 + \theta$. The bounds show that E_0 is the solution of $\tilde{E} = e_0$ with $e_0 \in [a_0 - \eta_1, a_0]$ and E_1 the solution of $\tilde{E} = e_1$ with $e_1 \geq a_0 + (1 - \epsilon)(a_1 - a_0) - \eta_1$; since $\tilde{E}$ has slope at most $1 + \theta$, $E_1 - E_0 \geq (e_1 - e_0)/(1 + \theta)$. $\square$

The constant shift costs nothing; its energy dependence costs the factor $1/(1 + \theta)$, and θ is the quantity that the cubic coupling makes large.

2. The fibered vacuum and its a -derivative on the abelian valley

On the abelian valley of the valley note, $a = \alpha T^3$, the charged modes at momentum k have energies $\hbar \omega_k^{(\pm)} = \hbar c |k \pm a|$ and the fibered vacuum is the product of Gaussians $\Omega_\perp(a) = \prod_k \Omega_k(a)$

with $\Omega_k \propto \exp[-\omega_k(a)|q_k|^2/(2\hbar \cdot \text{norm})]$ in the real mode coordinates q_k of each polarization and charge. For a single oscillator of frequency ω the derivative of the normalized ground state with respect to ω is $\partial_\omega \Omega = -\frac{1}{2\sqrt{2}\omega} \Omega_2$, with Ω_2 the normalized second excited state, so

$$\bar{P} \partial_{a_i} \Omega_\perp(a) = - \sum_{k, \pm, \text{pol}} \frac{\partial_{a_i} \omega_k^{(\pm)}}{2\sqrt{2} \omega_k^{(\pm)}} \Omega_{k,2} \otimes \prod_{k' \neq k} \Omega_{k'}, \quad \frac{\partial_{a_i} \omega_k^{(\pm)}}{\omega_k^{(\pm)}} = \pm \frac{(k \pm a)_i}{|k \pm a|^2}.$$

Its norm squared is $\sum_{k, \pm, \text{pol}} \frac{(k \pm a)_i^2}{8|k \pm a|^4} \simeq \frac{2 \cdot 2}{8} \sum_k \frac{\hat{k}_i^2}{|k|^2} = \frac{1}{2} \left(\frac{L}{2\pi}\right)^3 \frac{4\pi}{3} \Lambda [1 + O(a/\Lambda)] = \frac{L^3 \Lambda}{12\pi^2}$, linearly divergent: the vacua at different a are not close in norm as the cutoff is removed, which is the first sign that the fiber is ultraviolet-sensitive. What enters the Schur error is the same sum with the energy denominator, which converges better.

3. The Berry term

The constant-mode kinetic operator $\frac{g^2 c}{2\hbar L^3} \sum_i |\vec{p}_i|^2$ with $p_i = -i\hbar \partial_{a_i}$ acts on $\psi(a) \Omega_\perp(a)$ as $\frac{g^2 c}{2\hbar L^3} [(-\hbar^2 \Delta_a \psi) \Omega_\perp - 2\hbar^2 (\partial_{a_i} \psi) (\partial_{a_i} \Omega_\perp) - \hbar^2 \psi \Delta_a \Omega_\perp]$. The $\bar{P}$ -component of the middle term is the leading non-adiabatic coupling,

$$B_{\text{Berry}} \psi = -\frac{g^2 c \hbar}{L^3} \sum_i (\partial_{a_i} \psi) \bar{P} \partial_{a_i} \Omega_\perp.$$

Each two-quantum state $\Omega_{k,2}$ costs $D - E \geq 2\hbar \omega_k$ above the fiber, so

$$\|(D - E)^{-1/2} B_{\text{Berry}} \psi\|^2 \simeq \left(\frac{g^2 c \hbar}{L^3}\right)^2 \sum_i \sum_{k, \pm, \text{pol}} \frac{(k_i)^2}{8|k|^4 \cdot 2\hbar c |k|} |\partial_{a_i} \psi|^2 = \frac{g^4 c \hbar}{16L^6} \sum_i \left(\sum_k \frac{\hat{k}_i^2}{|k|^3}\right) |\partial_{a_i} \psi|^2,$$

with $\sum_k \hat{k}_i^2 |k|^{-3} \simeq \frac{1}{3} \left(\frac{L}{2\pi}\right)^3 4\pi \int_{2\pi/L}^{\Lambda} \frac{dk}{k} = \frac{L^3}{6\pi^2} \log \frac{\Lambda L}{2\pi}$. In the rescaled variable $a = g^{2/3} \xi / L$, $\partial_a = (L/g^{2/3}) \partial_\xi$, and

$$\eta_{\text{Berry}} \simeq \frac{g^4 c \hbar}{16L^6} \cdot \frac{L^3}{6\pi^2} \log(\Lambda L) \cdot \frac{L^2}{g^{4/3}} \langle |\partial_\xi \psi|^2 \rangle = \frac{g^{8/3}}{96\pi^2} \log(\Lambda L) \langle |\partial_\xi \psi|^2 \rangle \frac{\hbar c}{L}.$$

Since $\langle |\partial_\xi \psi|^2 \rangle$ is controlled by the kinetic part of h_3 , this is a relative bound of the form $R \leq \epsilon(A - a_0) + \eta_1$ with $\epsilon \sim g^2 \log(\Lambda L)/(96\pi^2)$. The logarithm is the ultraviolet signature of the vacuum polarization by the constant background; the one-loop coefficient of the running coupling for $SU(2)$ is $b_0 = 11/(24\pi^2)$, and the Berry term is one of the contributions that together must reproduce it, which is the consistency check the renormalized expansion has to pass. Its size, relative order $g^2 \log(\Lambda L)$, is what “ g is the renormalized coupling at scale L ” absorbs; at fixed cutoff it is small for $g^2 \log N_s \ll 1$.

4. The cubic coupling: constant shift and slope

$W_3 = \frac{\hbar c}{g^2} \int \bar{F}_{ij} \cdot [\tilde{A}_i, \tilde{A}_j]$ with $\tilde{A} = g \tilde{A}'$ creates three quanta from the fiber vacuum. With the mode normalization $\tilde{A}' \sim \sqrt{\hbar/(2\omega L^3)}(b + b^\dagger)$ the amplitude for (k, k', k'') , $k + k' + k'' = 0$, is of order $g \hbar c |k| (\hbar/(2\hbar c))^{3/2} L^{-3/2} (|k||k'||k''|)^{-1/2}$ times colour factors, and the second-order shift is

$$\eta_0 \sim g^2 \hbar c \sum_{k, k'} \frac{|k|^2/L^3}{|k||k'||k''| (|k| + |k'| + |k''|)} \sim g^2 \hbar c L^3 \Lambda^4 \cdot \text{const} = g^2 \frac{\hbar c}{L} (\Lambda L)^4 \cdot \text{const},$$

the two-loop vacuum energy of the box: extensive and quartically divergent, and independent of a at leading order. By Lemma 1' it costs nothing as a shift. Its energy derivative is

$$\theta = \eta'_0(E) \sim g^2 \sum_{k, k'} \frac{|k|^2/L^3}{|k||k'||k''| (|k| + |k'| + |k''|)^2} \cdot \frac{1}{\hbar c} \cdot \hbar c \sim g^2 (\Lambda L)^2 \cdot \text{const},$$

and Lemma 1' needs $\theta < 1$: the bare-fiber reduction is valid only for $g^2(\Lambda L)^2 \ll 1$. On a lattice $\Lambda L = \pi N_s$, so the condition is $g \ll 1/N_s$. The a -dependent part of the W_3 shift is the two-loop contribution to the valley potential, of order $g^2 \hbar c/L$ with a logarithm, and poses no separate problem.

5. What is provable and what is not

Fixed-lattice theorem (available by this route). On the periodic lattice with N_s sites per side, with the lattice regularization of the nonzero modes, for $g \leq g_1/N_s$ with g_1 a pure number, the hypotheses of Lemma 1' hold with $\epsilon, \theta = O(g^2 N_s^2)$ and $\eta_1 = O(g^{4/3} \hbar c/L)$, the compression is $h_3 + U$ with U the valley potential, and therefore

$$\Delta_{a,L} \geq \delta_1^{(3)} g^{2/3} \frac{\hbar c}{L} (1 - C g^{2/3} - C' g^2 N_s^2).$$

This is the statement that the zero-mode picture of C133 is exact at leading order on any fixed lattice as $g \rightarrow 0$: Lüscher's leading term at fixed regularization. Its proof is the bookkeeping of Sections 2–4 with rigorous constants; it is not carried out here.

Renormalized regime (not available by this route). The continuum small-volume expansion holds for $g(L)^2 \log(\Lambda L) \ll 1$, a far weaker condition than $g N_s \ll 1$. In that regime the bare Gaussian vacuum of the nonzero modes is not close to their true vacuum: the two-loop dressing η_0 exceeds the one-quantum gap $2\pi \hbar c/L$ by the factor $g^2 N_s^4$. A reduction that reaches the renormalized regime must use a dressed fiber, the ground state of the nonzero modes at $a = 0$ including their self-interaction, and treat only the a -dependence perturbatively; constructing that dressed vacuum with cutoff-uniform control is a renormalization theorem, the finite-volume ultraviolet problem that Balaban's work addresses on the Euclidean side.

Consequence. The small-volume corner S is not a finite-dimensional problem. Its lower side splits into a fixed-lattice theorem, provable by the present tools, and a renormalized theorem that is part of T4. This is consistent with the structure found in the obligations map: every statement uniform in the cutoff is a renormalization statement, and the zero-mode mechanism supplies the physics but not the ultraviolet control.

6. Consequence for STATE

Step 1 of the route (close the corner) becomes: (1a) the fixed-lattice theorem, $\Delta \geq \delta_1 g^{2/3} \hbar c/L (1 - o(1))$ for $g \ll 1/N_s$, provable now with explicit constants; (1b) the renormalized version, which requires a dressed nonzero-mode vacuum with cutoff-uniform bounds and is a T4-type problem. The Berry term's logarithm, with its coefficient, is the first check that a dressed reduction must pass against the one-loop b_0 . The blocking induction of the foreseen route inherits the same split: each blocking step is a fixed-cutoff statement, and its uniformity is the renormalization problem.